

Cybersecurity & CTF Study Plan

Beginner → Intermediate CTF Player | Path Toward an International Pentester Role

Your Starting Point

Comfortable with core Linux usage, file handling, and tools like strings, wget, curl, and chmod. Able to solve simple web challenges (robots.txt, hidden paths) and basic binary challenges with hints. Can run Python crypto scripts and follow the logic, but underlying concepts still need strengthening. Reverse engineering needs a more structured, repeatable approach.

The Goal

Build consistent intermediate-level CTF skills across web, binary/reverse engineering, and cryptography, then transition into practical pentesting methodology — working toward international pentester roles (target: ~60 LPA / ~\$72k range, realistically achievable with OSCP-level certification plus 2–4 years of demonstrated experience).

Weekly Time Commitment

Day Type	Hours	Focus
Monday – Friday	2 hrs/day	Focused, single-topic practice (deep work, not scattered)
Saturday	3–4 hrs	Extended lab/box work + review of the week
Sunday	3–4 hrs	New concept learning + write-ups + light rest

Total: ~16–18 hours/week. Consistency matters more than intensity — missing one day is fine, missing a whole week breaks momentum.

12-Week Plan Structure

The plan is split into 4 phases of 3 weeks each. Each phase has a clear goal, a recurring weekly rhythm, and a checkpoint to assess before moving on. Repeat or extend a phase if you're not yet consistent — this is normal and expected.

Phase	Weeks	Goal
1 — Foundations Reinforcement	1–3	Close gaps in core categories; finish PicoCTF; start OverTheWire
2 — Structured RE & Crypto Depth	4–6	Build repeatable reverse engineering workflow; understand crypto, not just run scripts
3 — Real-World Methodology	7–9	HackTheBox boxes; learn enumeration → exploitation → privesc methodology
4 — Certification Prep Launch	10–12	Begin structured OSCP-track prep; build a portfolio/write-up habit

Phase 1: Foundations Reinforcement (Weeks 1–3)

Goal: Close out PicoCTF fully, start OverTheWire Bandit/Natas/Krypton, and build the habit of writing a short note after every solved challenge explaining *why* it worked.

Weekly Rhythm (repeat for Weeks 1, 2, 3)

Day	Hrs	Focus Area	Tasks
Mon	2h	Web Exploitation	PicoCTF web challenges (continue from where you left off); note down any new technique used.
Tue	2h	Reverse Engineering	crackmes.one (easy tier) using fixed workflow: file → strings → Ghidra static read → gdb dynamic check.
Wed	2h	Cryptography	CryptoHack — Introduction + XOR sections. Write 2–3 sentences on why each attack works.
Thu	2h	Linux/OverTheWire	OverTheWire Bandit levels (aim for 3–5 levels/session); review any new command learned.
Fri	2h	Mixed CTF Practice	PicoCTF binary exploitation or forensics category; focus on categories you've avoided.
Sat	3–4h	Deep Practice Block	OverTheWire Natas (web) levels; if stuck >30 min on one level, look up a hint, then re-solve it without the hint the next day.
Sun	3–4h	Concept Building + Review	Read 1 write-up of a challenge you solved differently than the author. Watch 1 lppSec or LiveOverflow video relevant to a weak category. Update a personal notes doc.

Checkpoint (end of Week 3): You should have finished PicoCTF, completed Bandit, and be partway through Natas and Krypton. If reverse engineering still feels random rather than procedural, repeat Week 2–3's RE days before moving to Phase 2.

Phase 2: Structured RE & Crypto Depth (Weeks 4–6)

Goal: Make reverse engineering a repeatable process, not a guessing game. Understand the math/logic behind common crypto attacks (not just script execution).

Weekly Rhythm (repeat for Weeks 4, 5, 6)

Day	Hrs	Focus Area	Tasks
Mon	2h	Reverse Engineering	ROP Emporium challenges (start from level 0). Learn calling conventions, stack layout, and basic ROP chain construction.
Tue	2h	Cryptography Theory	CryptoHack — Mathematics + RSA sections. Implement one attack (e.g. small-exponent RSA) by hand in Python, not copy-pasted.
Wed	2h	Web Exploitation	TryHackMe Jr Penetration Tester path — web-focused rooms (SQLi, IDOR, auth bypass fundamentals).
Thu	2h	Reverse Engineering	pwn.college intro modules or crackmes.one medium tier; practice writing a short methodology note per binary.
Fri	2h	Cryptography	OverTheWire Krypton levels; for each, identify the cipher type before looking at any solution.
Sat	3–4h	HTB Starting Point	Begin HackTheBox Starting Point (Tier 0/1 machines). Document enumeration steps for every box, even 'easy' ones.
Sun	3–4h	Tooling + Review	Set up and learn Burp Suite properly (repeater, intruder basics). Review the week's hardest challenge and re-solve from scratch.

Checkpoint (end of Week 6): You should be able to look at an unfamiliar binary and know the first 3 steps to take without hints. You should be able to explain RSA's basic vulnerability classes out loud, not just run an attack script.

Phase 3: Real-World Methodology (Weeks 7–9)

Goal: Shift from isolated CTF puzzles to full pentest methodology — enumeration, exploitation, privilege escalation, and reporting, the way it's actually done on the job.

Weekly Rhythm (repeat for Weeks 7, 8, 9)

Day	Hrs	Focus Area	Tasks
Mon	2h	HTB Easy Boxes	Pick 1 Easy-rated Linux box. Full enumeration (nmap, gobuster/feroxbuster) before touching exploitation.
Tue	2h	HTB Easy Boxes (cont.)	Continue same box or move to exploitation phase; document each step as if writing a report.
Wed	2h	Privilege Escalation	Study Linux privesc techniques (GTFOBins, misconfigured SUID, cron jobs) using a dedicated privesc practice VM.
Thu	2h	HTB Windows Box	Pick 1 Easy-rated Windows box; learn basic AD enumeration concepts if applicable.
Fri	2h	Report Writing Practice	Write a mini pentest report for a box solved this week (target, steps, screenshots, remediation) — this becomes portfolio material.
Sat	3–4h	New HTB Box (Full Cycle)	Full box solo, start to finish, no walkthroughs unless stuck 45+ minutes.
Sun	3–4h	Review + AD Basics Intro	Watch IppSec walkthrough of a box you solved — compare your method to theirs. Start light reading on Active Directory concepts (groundwork for OSCP/CRTF later).

Checkpoint (end of Week 9): You should be solving HTB Easy boxes with under ~30% reliance on walkthroughs, and have 3–4 mini reports written. This report-writing habit matters a lot for real pentest job interviews.

Phase 4: Certification Prep Launch (Weeks 10–12)

Goal: Begin structured prep toward OSCP (or eJPT first if you want an earlier milestone), and build a public portfolio (write-ups, GitHub, LinkedIn activity) that recruiters can actually see.

Weekly Rhythm (repeat for Weeks 10, 11, 12)

Day	Hrs	Focus Area	Tasks
Mon	2h	OSCP-style Practice	HTB or TJNull's OSCP-like box list (Medium difficulty); focus on methodology consistency.
Tue	2h	OSCP-style Practice	Continue same box or new one; practice working without metasploit where possible (manual exploitation builds deeper understanding).
Wed	2h	Active Directory	TryHackMe or HTB AD-focused rooms; learn enumeration tools (BloodHound basics, enum4linux).
Thu	2h	Portfolio Building	Publish a clean write-up (sanitized) on GitHub/blog for a box or CTF challenge solved this phase.
Fri	2h	Mixed Review	Revisit 1 challenge from each category (web/RE/crypto) from earlier phases — confirm retention.
Sat	3–4h	Full Box Simulation	Timed full-box attempt (treat it like exam conditions: enumerate, exploit, privesc, document, all in one sitting).
Sun	3–4h	Career Prep	Update LinkedIn/resume with new write-ups and skills. Research specific international pentester job postings to track what they actually ask for.

Checkpoint (end of Week 12): Decide whether to enroll in OSCP now or extend practice for another 4–6 weeks. There's no rush — going in under-prepared wastes both money and confidence.

Key Resources Reference

Category	Platform	Purpose
All-round CTF	PicoCTF	Beginner-friendly, structured difficulty curve
Linux + Web	OverTheWire (Bandit, Natas, Krypton)	Linux fundamentals, web exploitation, crypto basics
Guided Learning Paths	TryHackMe	Jr Penetration Tester path, AD basics
Real-world Pentest Practice	HackTheBox	Starting Point → Easy boxes → OSCP-like list
Cryptography Depth	CryptoHack	Conceptual understanding, not just script execution
Reverse Engineering	crackmes.one, ROP Emporium, pwn.college	Structured RE and binary exploitation workflow
Walkthrough Learning	IppSec (YouTube), LiveOverflow (YouTube)	Compare your methodology to experts after solving
Certification Target	OSCP (primary), eJPT (stepping stone), CRTP (AD-focused)	Internationally recognized credentials for pentest roles

Honest Notes on the Career Goal

A ~60 LPA (~\$72k USD) international pentester role is realistic, but typically aligns with 2–4+ years of experience plus a recognized certification like OSCP, rather than an entry-level hire. A practical path: build skills via this plan → earn OSCP → spend 1–2 years in a SOC or junior pentest role (your SIEM/SOC background is a real advantage here) → then target international roles with a track record, certification, and portfolio backing the application.

Review this plan every 3 weeks at each checkpoint. If a phase feels rushed, extend it — real retention matters more than calendar speed.